

Problem Set 1

1

You're an engineer working on the smart toilet and your job is to figure out how deep the bowl should be. You are also planning to add some kind of shock absorber on the wall to cancel out the returning ripple, but it must fire at exactly the right time.

After experimenting you learn that when an "item" lands in the water, it generates a ripple following the equation:

$$y(x, t) = 6 \sin(\pi(4t - 0.2x)) \text{ (units: cm, s)}$$

a

How high will the ripples reach above the still water surface?

✓ Answer ✓

The height of the $y(x, t)$ function will be 6 *cm* as that is the max of that function.

b

How long will it take for the ripples to reach the bowl edge if the bowl is 40 cm in diameter?

✓ Answer

If we drop it in the center (0 cm) at time 0s, we evaluate the $\sin$ insides as 0.

The ripple will reach the edge when that inside value reaches the same evaluation at the edge (20 cm).

$$4t - 0.2(20) = 0$$

$$\implies t = 1 \text{ s}$$

2

Jupiter is about 900 million kilometers from Earth. A probe is traveling through space towards Jupiter and periodically emits a 1 *GHz* radio pulse while listening for the return echo. The time to the echo tells the probe how soon it will crash into Jupiter.

a

Right after leaving Earth, how long will it take to receive an echo from Jupiter?

✓ Answer

$$c \approx 3.0 \times 10^8 \text{ m/s}$$

$$d = 9.0 \times 10^{11} \text{ m}$$

$$t = \frac{d}{c} = 3.0 \times 10^3 \text{ s}$$

$$2t = \frac{d}{c} = 6.0 \times 10^3 \text{ s}$$

$$= 1 \text{ hour, 40 mins}$$

b

The probe is slowing down but still going 1 km/s and receives an echo after $66.6 \mu\text{s}$. How soon until it hits Jupiter?

✓ **Answer**

$$v = 1 \text{ km/s}$$

$$t = \frac{2}{3} \times 10^{-4} \text{ s}$$

$$d = \frac{vt}{2} = 10 \text{ km}$$

$$t = \frac{d}{v} = 10 \text{ s}$$

1.4

A wave traveling along a string is given by

$$y(x, t) = 2 \sin(4\pi t + 10\pi x) \text{ cm}$$

Where x is the distance along the string in meters and y is the vertical displacement.

a

Determine the direction of the wave travel

✓ **Answer**

To the left as there is a $+$ component.

b

Determine the reference phase ϕ_0

✓ **Answer**

$\phi_0 = 0$ as there is no constant in the trig function.

c

Determine the frequency

✓ Answer

$$f = \frac{4\pi}{2\pi} = 2 \text{ Hz}$$

d

Determine the wavelength

✓ Answer

$$\frac{2\pi}{10\pi} = 0.2 \text{ cm}$$

e

Determine the phase velocity

✓ Answer

$$\frac{-4}{10} = -0.4 \text{ cm/s}$$

1.8

Two waves on a string are given by the functions:

$$y_1(x, t) = 4 \cos(20t - 30x) \text{ cm}$$

$$y_2(x, t) = -4 \cos(20t + 30x) \text{ cm}$$

Where x is in centimeters. The waves are said to interfere constructively when their superposition $|y_s| = |y_1 + y_2|$ is a maximum, and they interfere destructively when $|y_s|$ is a minimum.

a

What are the directions of propagation of waves $y_1(x, t)$ and $y_2(x, t)$?

✓ Answer

Positive, and negative.

b

At $t = (\frac{\pi}{50}) \text{ s}$, what location x do the two waves interfere constructively, and what is the corresponding value of $|y_s|$?

✓ Answer

$$y_s = 4(\cos(20t - 30x) - \cos(20t + 30x))$$

$$y_s = 8(\sin(20t) \sin(30x))$$

$$y_s = 8\left(\sin\left(\frac{2\pi}{5}\right) \sin(30x)\right)$$

$$x = \frac{\pi}{60} \text{ cm}$$

$$y_s = 7.60845 \text{ cm}$$

c

At $t = (\frac{\pi}{50}) \text{ s}$, what location x do the two waves interfere destructively, and what is the corresponding value of $|y_s|$?

✓ **Answer**

$$y_s = 8 \left(\sin\left(\frac{2\pi}{5}\right) \sin(30x) \right)$$

$$x = 0 \text{ cm}$$

$$y_s = 0 \text{ cm}$$

1.19

Complex numbers z_1 and z_2 are given by

$$z_1 = -3 + j2$$

$$z_2 = 1 - j2$$

a

Determine $z_1 z_2$

✓ **Answer**

$$1 - j4$$

b

Determine $\frac{z_1}{z_2^*}$

✓ **Answer**

$$(-3 + j2)/(1 + j2) = \frac{1}{5} + j\frac{8}{5}$$

c

Determine z_1^2

✓ **Answer**

$$5 - j12$$

d

Determine $z_1 z_1^*$

✓ Answer

13

1.25

If $z = 3 - j4$, find the value of e^z

✓ Answer

e^{3-j4}

$e^3(\cos 4 - j \sin 4)$